Lecture 3: Recursive Functions

Models of Computation

https://clegra.github.io/moc.html

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Ph.D. Program, Advanced Courses Period

Gran Sasso Science Institute

L'Aquila, Italy

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Course overview

Monday, July 6 10.30 – 12.30	Tuesday, July 7 10.30 – 12.30	Wednesday, July 8 10.30 – 12.30	Thursday, July 9 10.30 – 12.30	Friday, July 10
intro	*classic models*			*additional models*
Introduction to Computability	**Machine Models**	**Recursive Functions**	**Lambda Calculus**	
computation and decision problems, from logic to computability, overview of models of computation relevance of MoCs	Post Machines, typical features, Turing's analysis of human computers, Turing machines, basic recursion theory	primitive recursive functions, Gödel–Herbrand recursive functions, partial recursive funct's, partial recursive = = Turing-computable, Church's Thesis	λ-terms, β-reduction, λ-definable functions, partial recursive = λ-definable = Turing computable	
	imperative programming	*algebraic programming*	*functional programming*	
				14.30 – 16.30
				Three more Models of Computation
				Post's Correspondence Problem, Interaction-Nets, Fractran
				comparing computational power

Overview

Turing machines / elementary recursion theory

- Chomsky hierarchy
- Exercises Turing machines
- Halting Problem

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  - defined with μ-recursion (unbounded minimisation)

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  - effectively calculable functions $\stackrel{\wedge}{=}$ partial-recursive functions

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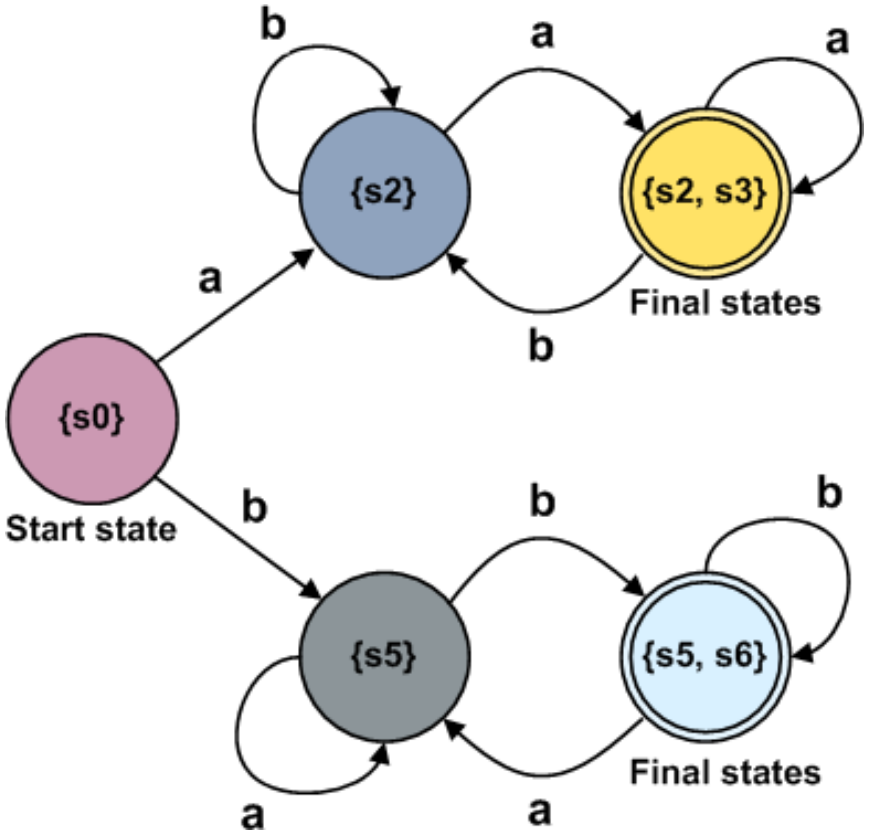
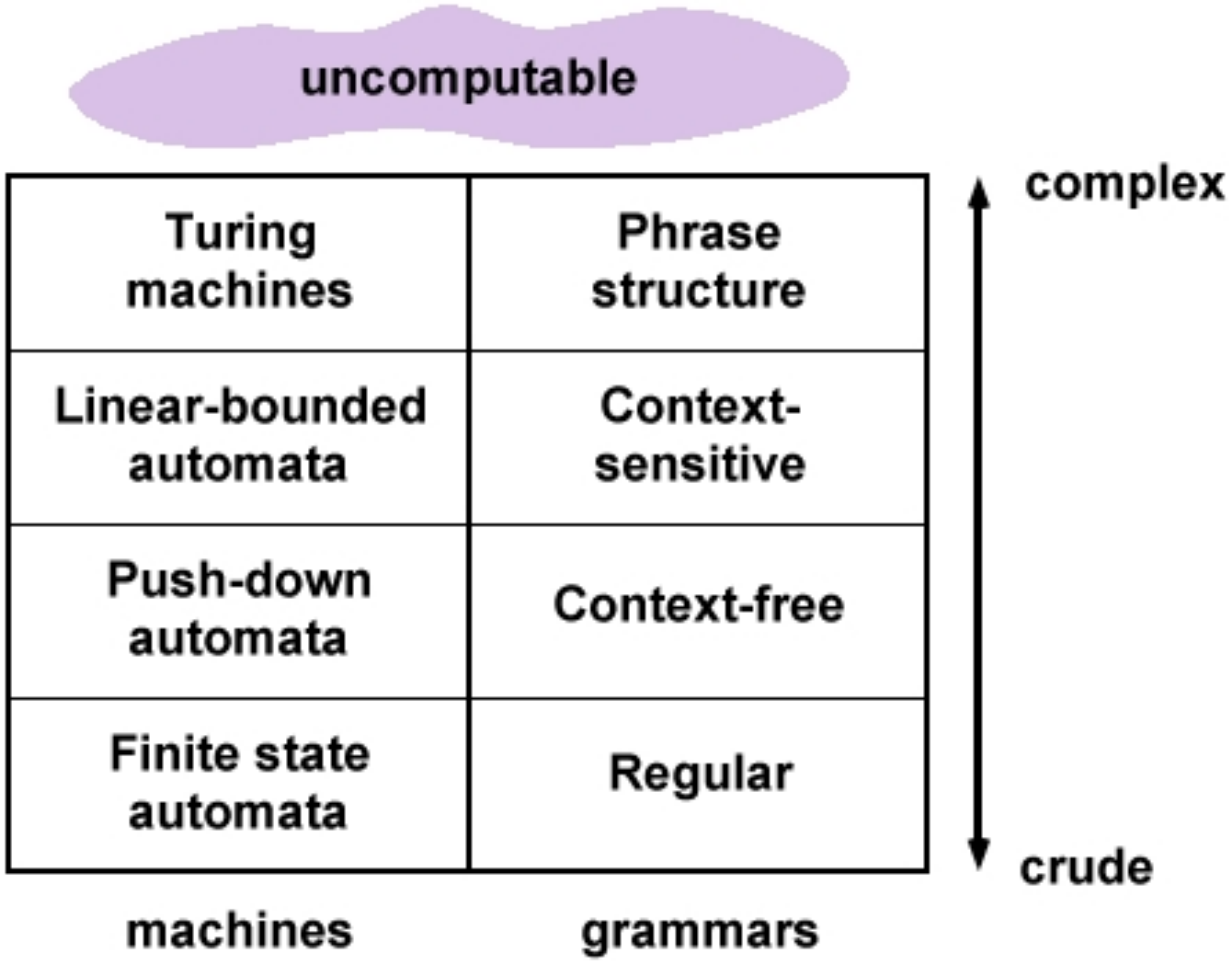
- Partial recursive functions = Turing computable functions
- Church's thesis
  - effectively calculable functions $\stackrel{\wedge}{=}$ partial-recursive functions
  - some debate

Finite-state automaton



Formal-languages Chomsky hierarchy



Exercises

Exercise

Construct a Turing machine that adds one to a natural number in binary representation.

(In the film this Turing machine is executed five times consecutively.)

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Exercise

Construct a Turing machine that, if started on the empty tape, writes the sequence

$$0101101110111101111 0\ldots$$

on the tape, but does not halt.

(Compare your machine with Turing's machine for this purpose.)

Exercise: Halting Problem

Exercise

Try to adapt the diagonalisation argument to show that for the Halting Problem

$$H = \{ \langle \langle M \rangle, w \rangle \mid M \text{ halts on input } w \}$$

it holds:

- H is not recursive

and show that:

- H is recursively enumerable

Timeline: From logic to computability

1900	Hilbert's 23 Problems in mathematics
1910/12/13	Russell/Whitehead: Principia Mathematica
1928	Hilbert/Ackermann: formulate completeness/decision problems for the predicate calculus (the latter called 'Entscheidungsproblem')
1929	Presburger: completeness/decidability of theory of addition on $\mathbb{Z}$
1930	Gödel: completeness theorem of predicate calculus
1931	Gödel: incompleteness theorems for first-order arithmetic
1932	Church: λ-calculus
1933/34	Herbrand/Gödel: general recursive functions
1936	Church/Kleene: λ-definable $\sim$ general recursive Church Thesis: 'effectively calculable' be defined as either Church shows: the 'Entscheidungsproblem' is unsolvable Post: machine model; Church's thesis as 'working hypothesis'
1937	Turing: convincing analysis of a 'human computer' leading to the 'Turing machine'

Busy Beaver

Definition (Tibor Radó, 1962)

$BB(n)$:= the largest number of steps any n-state Turing machine with tape alphabet $\{0, 1\}$ and blank symbol 0 can run before eventually halting (counting halting as a step), when started on an empty tape.

$BB(1) = 1$ (easy)

$BB(2) = 6$ (homework exercise)

$BB(3) = 21$ (S. Lin and T. Radó, 1965)

$BB(4) = 107$ (A. Brady, 1983)

$BB(5) = \begin{cases} \geq 47176870 \\ = 47176870 \end{cases}$ (1990) (international team, 2024)

$BB(6) \geq \underbrace{10^{10^{\cdot^{\cdot^{\cdot^{10}}}}}}_{15}$

- Scott Aaronson: "The Busy Beaver Frontier", [1].
- —"Why Philosophers Should Care About Computational Complexity?" [2]

Busy Beaver is not computable

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Proposition

The busy beaver function $BB : \mathbb{N} \to \mathbb{N},\ n \mapsto BB(n)$ is not (Turing-)computable.
More generally: one can solve the Halting Problem, given oracle access to any function $b : \mathbb{N} \to \mathbb{N}$ such that $b(n) \geq BB(n)$ for all $n \in \mathbb{N}$.

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Proposition

Let $f : \mathbb{N} \to \mathbb{N}$ be a Turing-computable function.
Then there exists an $n_f \in \mathbb{N}$ such that $BB(n) > f(n)$ for all $n \geq n_f$.

Recursive Functions

Functions defined by recursive equations:

like e.g. functions $+, \cdot, (\cdot)^{\cdot} : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, and $(\cdot)! : \mathbb{N} \to \mathbb{N}$:

$$n + 0 = n$$
$$n + (m + 1) = (n + m) + 1$$

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Primitive recursive functions: defined by such equations (termination of the evaluation process guaranteed)

General recursive functions: defined by more general systems of equations

μ-Recursive (partial recursive) functions:

extend the primitive recursive functions by a μ-operator

that allows to obtain partial functions

Rósza Péter

Rósza Péter (1905–1977)

Primitive recursive functions ($\mathbb{N}^k \to \mathbb{N}$)

Base functions:

- $\mathcal{O} : \mathbb{N}^0 = \{\varnothing\} \to \mathbb{N}$, $\varnothing \mapsto 0$ (0-ary constant-0 function)
- succ : $\mathbb{N} \to \mathbb{N}$, $x \mapsto x + 1$ (successor function)
- $\pi_i^n : \mathbb{N}^n \to \mathbb{N}$, $\vec{x} = \langle x_1, \ldots, x_n \rangle \mapsto x_i$ (projection function)

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Closed under operations:

- composition: if $f : \mathbb{N}^k \to \mathbb{N}$, and $g_i : \mathbb{N}^n \to \mathbb{N}$ are prim. rec., then so is $h = f \circ (g_1 \times \ldots \times g_k) : \mathbb{N}^n \to \mathbb{N}$:

$$h(\vec{x}) = f(g_1(\vec{x}), \ldots, g_k(\vec{x}))$$

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- primitive recursion: if $f : \mathbb{N}^n \to \mathbb{N}$, $g : \mathbb{N}^{n+2} \to \mathbb{N}$ are prim. rec., then so is $h = \mathrm{pr}(f; g) : \mathbb{N}^{n+1} \to \mathbb{N}$:

$$h(\vec{x}, 0) = f(\vec{x})$$
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- $\pi_i^n : \mathbb{N}^n \to \mathbb{N}$, $\vec{x} = \langle x_1, \ldots, x_n \rangle \mapsto x_i$ (projection function)
- for $n > 1$: $\text{id}^n : \mathbb{N}^n \to \mathbb{N}^n$, $\vec{x} = \langle x_1, \ldots, x_n \rangle \mapsto \vec{x}$ (n-ary identity f.)

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Primitive recursive functions (exercises)

Exercise

Show that the following functions are primitive recursive:

- addition
- constant functions
- multiplication
- (positive) sign-function
- the representing functions $\chi_=$ and $\chi_<$ for the predicates $=$ and $<$.

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Try-yourself-Examples

Show that the following functions are primitive recursive:

- exponentiation
- factorial

Admissible operations for primitive recursive functions

Proposition

1. *definition by case distinction:*

$$f(\vec{x}) := \begin{cases} f_1(\vec{x}) & \dots P_1(\vec{x}) \\ f_2(\vec{x}) & \dots P_2(\vec{x}) \wedge \neg P_1(\vec{x}) \\ \dots & \\ f_k(\vec{x}) & \dots P_k(\vec{x}) \wedge \neg P_{k-1}(\vec{x}) \wedge \dots \wedge \neg P_1(\vec{x}) \\ f_{k+1}(\vec{x}) & \dots \neg P_k(\vec{x}) \wedge \dots \wedge \neg P_1(\vec{x}) \end{cases}$$

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2. *definition by bounded recursion:*

$$\mu z_{\leq y}.\ [P(x_1, \dots, x_n, z)] := \begin{cases} z & \dots \neg P(x_1, \dots, x_n, i) \text{ for } 0 \leq i < z \leq y, \\ & \quad \text{and } P(x_1, \dots, x_n, z) \\ y+1 & \dots \neg \exists z.\big(0 \leq z \leq y \wedge P(x_1, \dots, x_n, z)\big) \end{cases}$$

Properties of primitive recursive functions

Proposition

1. *Every primitive recursive function is total.*

Properties of primitive recursive functions

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1. *Every primitive recursive function is total.*
2. *Every primitive recursive function is Turing-computable.*

Properties of primitive recursive functions

Definition

A total function $f : \mathbb{N}^k \to \mathbb{N}$ is Turing-computable if there exists a Turing machine $M = \langle Q, \Sigma, \Gamma, \delta, q_0, \textbb, F \rangle$ and a calculable coding function $\langle \cdot \rangle : \mathbb{N} \to \Sigma^*$ such that:

- for all $n_1, \ldots, n_k \in \mathbb{N}$ there exists $q \in F$ such that:

$$q_0 \langle n_1 \rangle \textbb \langle n_2 \rangle \textbb \ldots \textbb \langle n_k \rangle \vdash_M^* q \langle f(n_1, \ldots, n_k) \rangle$$

Properties of primitive recursive functions

Proposition

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Proof.

For (2):

- the base functions are Turing-computable
- the Turing-computible functions are closed under the schemes composition and primitive recursion

□

Features of computationally complete MoC's present?

- storage (unbounded)
- control (finite, given)
- modification
  - of (immediately accessible) stored data
  - of control state
- conditionals
- loop (unbounded)
- stopping condition

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Not primitive recursive (I)

Proposition

There exist calculable/Turing-computable functions that are not primitive recursive.

Proof.

By diagonalisation. □

Not primitive recursive (II): Ackermann function

Wilhelm Ackermann (1896–1962)

Not primitive recursive (II): Ackermann function

Ackermann function $\mathsf{A} : \mathbb{N}^2 \to \mathbb{N}$ (simplified version by Rósza Péter):

$$\mathsf{A}(0, x) = \mathsf{Succ}(x)$$

$$\mathsf{A}(x + 1, 0) = \mathsf{A}(x, \mathsf{Succ}(0))$$

$$\mathsf{A}(x + 1, y + 1) = \mathsf{A}(x, \mathsf{A}(x + 1, y))$$

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A is not primitive recursive, it grows too fast:

$$A(0, n) = n + 1$$
$$A(1, n) = n + 2$$
$$A(2, n) = 2n + 3$$
$$A(3, n) = 2^{n+3} - 2$$
$$A(4, n) = \underbrace{2^{2^{\cdot^{\cdot^{\cdot^{2^{16}}}}}}}_{n} - 3$$
$$\ldots$$

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A grows faster than every primitive recursive function:

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A grows faster than every primitive recursive function:

Theorem

For every primitive recursive $f : \mathbb{N} \to \mathbb{N}$ there exists some $i \in \mathbb{N}$ such that $f(i) < \mathsf{A}(i, i)$.

Not primitive recursive (II): Ackermann function

Ackermann function $\mathsf{A} : \mathbb{N}^2 \to \mathbb{N}$ (simplified version by Rósza Péter):

$$\mathsf{A}(0, y) = \mathsf{Succ}(y)$$

$$\mathsf{A}(x + 1, 0) = \mathsf{A}(x, \mathsf{Succ}(0))$$

$$\mathsf{A}(x + 1, y + 1) = \mathsf{A}(x, \mathsf{A}(x + 1, y))$$

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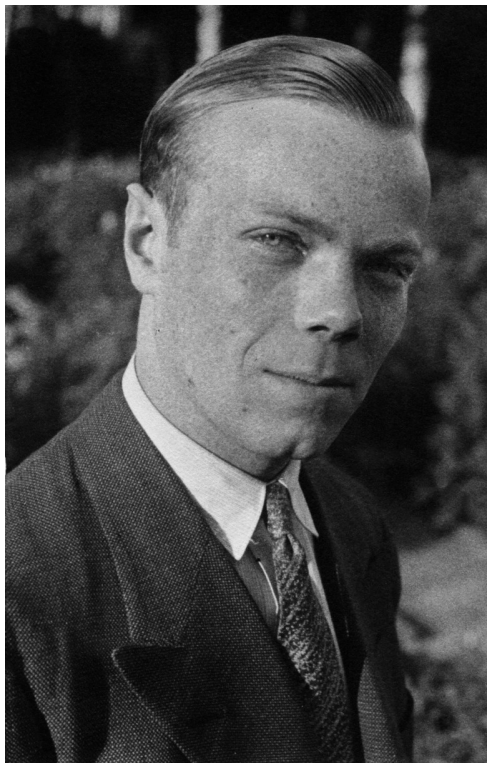
Theorem (Green, 1964)

For all $n \in \mathbb{N}$:

$$BB(2n) \geq \mathsf{A}(n, n)\,.$$

Jacques Herbrand



Jacques Herbrand (1908–1931)

Kurt Gödel

Kurt Gödel (1906–1978)

Gödel–Herbrand general recursive function

Defined by systems of recursion equations like that for the Ackermann function:

$$\mathsf{A}(0, y) = \mathsf{Succ}(y)$$

$$\mathsf{A}(\mathsf{Succ}(x), 0) = \mathsf{A}(x, \mathsf{Succ}(0))$$

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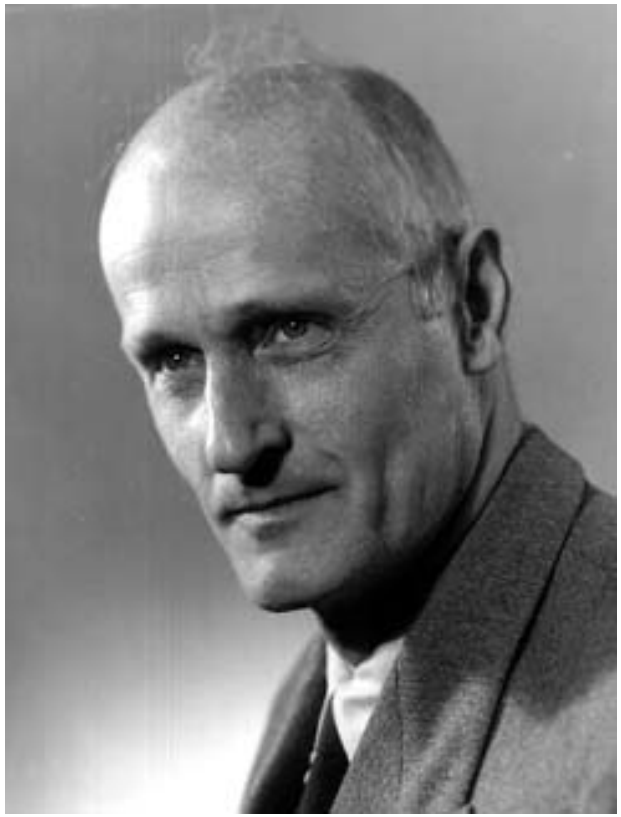
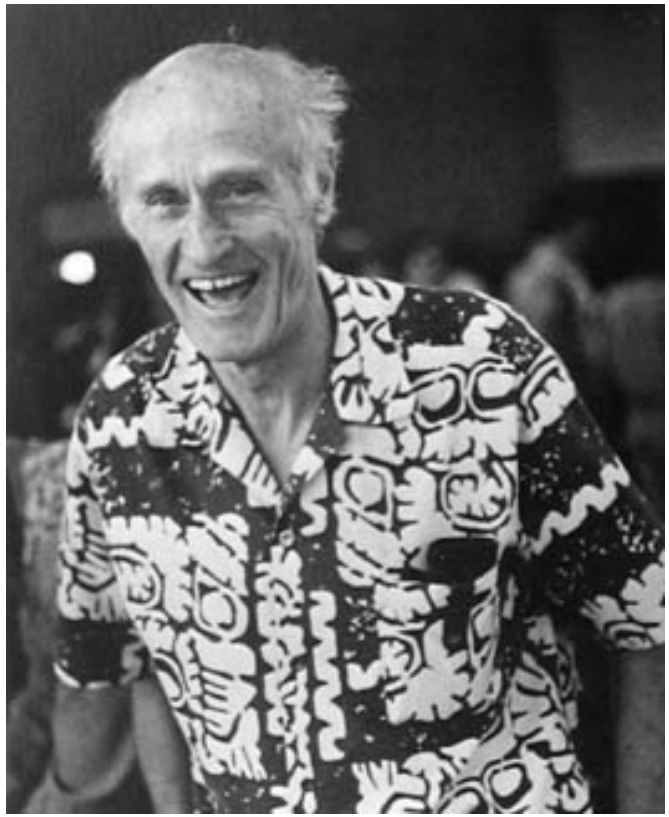
Numerals: $\langle 0 \rangle := 0$, and $\langle n \rangle := \underbrace{\mathsf{Succ}(\ldots \mathsf{Succ}(0))}_{n}$ for $n > 1$.

Definition

A function $f : \mathbb{N}^k \to \mathbb{N}$ is called general recursive if it can be defined by (such a) system $\mathcal{S}$ of recursion equations via a function symbol F if for all $n_1, \ldots, n_k \in \mathbb{N}$, the expression $F(\langle n_1 \rangle, \ldots, \langle n_k \rangle)$ evaluates according to $\mathcal{S}$ to a unique numeral $\langle n \rangle$, and such that furthermore: $n = f(n_1, \ldots, n_k)$.

Stephen Cole Kleene





Stephen Cole Kleene (1906–1994)

Unbounded minimisation (μ-recursion)

Let $f : \mathbb{N}^{k+1} \to \mathbb{N}$ total. Then the partial function defined by:

$$\mu(f) : \mathbb{N}^k \rightharpoonup \mathbb{N}$$

$$\vec{x} \mapsto \begin{cases} \min\{y \in \mathbb{N} \mid f(\vec{x}, y) = 0\} & \ldots \exists y\,(f(\vec{x}, y) = 0) \\ \uparrow & \ldots \text{else} \end{cases}$$

is called the unbounded minimisation of f.

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Let $f : \mathbb{N}^{k+1} \rightharpoonup \mathbb{N}$ partial. Then the partial function $\mu(f)$:

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$$\vec{x} \mapsto \begin{cases} z & \ldots f(\vec{x}, z) = 0 \ \wedge\ \forall y\,\big(0 \le y < z \to (f(\vec{x}, y)\downarrow \neq 0)\big) \\ \uparrow & \ldots \neg\exists y\,\big(f(\vec{x}, y) = 0 \ \wedge\ \forall z\,(0 \le z < y \to (f(\vec{x}, z)\downarrow)\big) \end{cases}$$

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Partial, and total, recursive functions

Definition

A partial function $f : \mathbb{N}^n \rightharpoonup \mathbb{N}^l$ is called partial recursive if it can be specified from base functions ($\mathcal{O}$, succ, π_i^n, and id^n) by successive applications of composition, primitive recursion, and unbounded minimisation.

A partial recursive function is called (total) recursive if it is total.

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Proposition

Every partial recursive function is Turing-computable.

Primitive recursive

- storage (unbounded) ✓
- control (finite, given) ✓
- modification ✓
  - of (immediately accessible) stored data
  - of control state
- conditionals ✓
- loop ✓ (unbounded) ×
- stopping condition ✓

Partial recursive = prim. rec. + unbounded minimization

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Turing-computable functions

Definition

1. A total function $f : \mathbb{N}^k \to \mathbb{N}$ is Turing-computable if there exists a Turing machine $M = \langle Q, \Sigma, \Gamma, \delta, q_0, \not b, F \rangle$ and a calculable coding function $\langle \cdot \rangle : \mathbb{N} \to \Sigma^*$ such that:
   - for all $n_1, \ldots, n_k \in \mathbb{N}$ there exists $q \in F$ such that:

$$q_0 \langle n_1 \rangle \not b \langle n_2 \rangle \not b \ldots \not b \langle n_k \rangle \vdash_M^* q \langle f(n_1, \ldots, n_k) \rangle$$

Turing-computable functions

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Partial recursive vs. Turing-computable functions

Lemma

Every Turing-computable function is partial recursive.

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Proof by arithmetization of Turing machines, showing:

Theorem (Kleene's normal form theorem)

For every Turing-computable, partial function (and hence for every partial recursive function) $h : \mathbb{N}^k \to \mathbb{N}$ *there exist primitive recursive functions* $f : \mathbb{N} \to \mathbb{N}$ *and* $g : \mathbb{N}^{k+1} \to \mathbb{N}$ *such that:*

$$h(x_1, \ldots, x_n) = (f \circ \mu(g))(x_1, \ldots, x_n)$$

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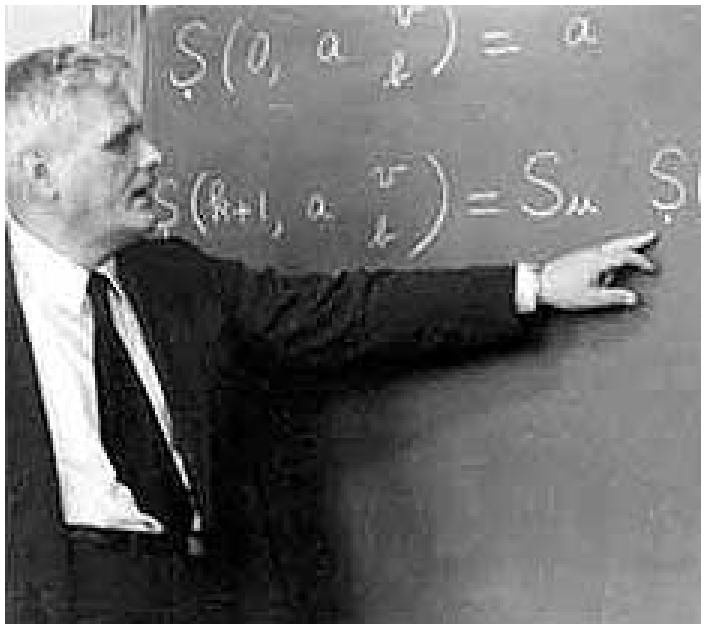
Theorem

The Turing-computable (partial) functions coincide with the partial recursive functions.

Alonzo Church



Alonzo Church (1903–1995)

Effectively calculable functions

Alonzo Church (1936):

> *"We now define the notion [. . .] of an effectively calculable function of positive integers by identifying it with the notion of a recursive function of positive integers (or a λ-definable function of positive integers). This definition is thought to be justified by the considerations which follow, so far as positive justification can ever be obtained for the selection of formal definition to correspond to an intuitive notion."*

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Definition (Church)

For every total function $f : \mathbb{N} \to \mathbb{N}$, and partial function $g : \mathbb{N} \rightharpoonup \mathbb{N}$,

$$f \text{ is effectively calculable} \iff f \text{ is recursive}$$
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Church's Thesis

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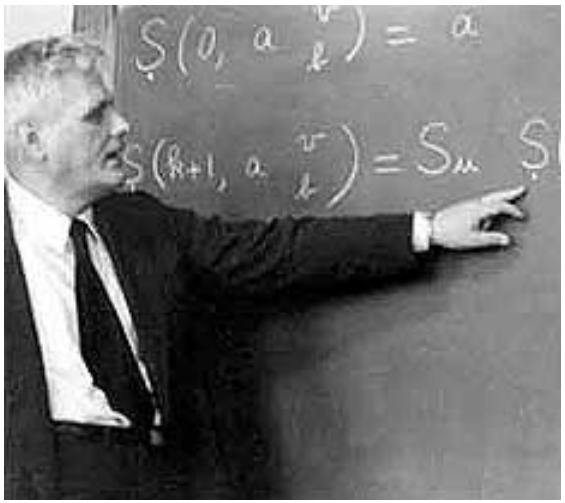
λ-calculus



Alonzo Church (1903 –1992)

Theorem (Kleene/Church, 1935)

Every λ-definable function is general recursive, and vice versa.

Recommended reading

1. Recursive and primitive-recursive functions:

   Chapter 3, The Lambda Calculus of the book:

   - Maribel Fernández [4]: *Models of Computation (An Introduction to Computability Theory)*, Springer-Verlag London, 2009.

Post's 'working hypothesis'

E.L. Post in his 1936 article (Post machines):

> *"The writer expects the present formulation to turn out to be logically equivalent to recursiveness in the sense of the Gödel–Church development. Its purpose, however, is not only to present a system of a certain logical potency but also, in its restricted field, of psychological fidelity. In the latter sense wider and wider formulations are contemplated. On the other hand, our aim will be to show that all such are logically reducible to formulation 1 [Post machines]. We offer this conclusion at the present moment as a working hypothesis. And to our mind such is Church's identification of effective calculability with recursiveness."*

Church on Post's 'working hypothesis'

Alonzo Church in his review (1937) of Post's 1936 article:

> *"The author proposes a definition of "finite 1-process" which is similar in formulation, and in fact equivalent, to computation by a Turing machine (see the preceding review). He does not, however, regard his formulation as certainly to be identified with effectiveness in the ordinary sense, but takes this identification as a "working hypothesis" in need of continual verification. To this the reviewer would object that effectiveness in the ordinary sense has not been given an exact definition, and hence the working hypothesis in question has not an exact meaning. To define effectiveness as computability by an arbitrary machine, subject to restrictions of finiteness, would seem to be an adequate representation of the ordinary notion, and if this is done the need for a working hypothesis disappears."*

Church on Turing's paper

A. Church in his review (1937) of Turing's 1936 article:

> *"The author proposes as a criterion that an infinite sequence of digits 0 and 1 be "computable" that it shall be possible to devise a computing machine, occupying a finite space and with working parts of finite size, which will write down the sequence to any desired number of terms if allowed to run for a sufficiently long time. As a matter of convenience, certain further restrictions are imposed on the character of the machine, but these are of such a nature as obviously to cause no loss of generality—in particular, a human calculator, provided with pencil and paper and explicit instructions, can be regarded as a kind of Turing machine. It is thus immediately clear that computability, so defined, can be identified with (especially, is no less general than) the notion of effectiveness as it appears in certain mathematical problems [. . .].*

Busy Beaver and axiomatic-theory consistency

Proposition

Let T be a computable and arithmetically sound axiomatic theory. Then there exists a constant n_T such that for all $n \geq n_T$, no statement of the form "BB(n) = k" can be proved in T.

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There is an explizit 27-state Turing machine that halts iff Goldbach's Conjecture is false.

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Theorem (Matiyasevich, O'Rear, Aaronson)

There is an explicit 744-state Turing machine that halts iff the Riemann Hypothesis is false.

Summary

Recursive functions

- primitive recursive functions
- Gödel–Herbrand(–Kleene) general recursive functions
- partial recursive functions
  - defined with μ-recursion (unbounded minimisation)
- Partial recursive functions = Turing computable functions
- Church's thesis
  - effectively calculable functions $\stackrel{\wedge}{=}$ partial-recursive functions
  - some debate

Course overview

Monday, July 6 10.30 – 12.30	Tuesday, July 7 10.30 – 12.30	Wednesday, July 8 10.30 – 12.30	Thursday, July 9 10.30 – 12.30	Friday, July 10
intro	*classic models*			*additional models*
Introduction to Computability	**Machine Models**	**Recursive Functions**	**Lambda Calculus**	
computation and decision problems, from logic to computability, overview of models of computation relevance of MoCs	Post Machines, typical features, Turing's analysis of human computers, Turing machines, basic recursion theory	primitive recursive functions, Gödel–Herbrand recursive functions, partial recursive funct's, partial recursive = = Turing-computable, Church's Thesis	λ-terms, β-reduction, λ-definable functions, partial recursive = λ-definable = Turing computable	
	imperative programming	*algebraic programming*	*functional programming*	
				14.30 – 16.30
				Three more Models of Computation
				Post's Correspondence Problem, Interaction-Nets, Fractran
				comparing computational power

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on youtube at `https://youtu.be/OST1DjD08Hg?si=PS9O0x3szKF7LhSZ`.

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